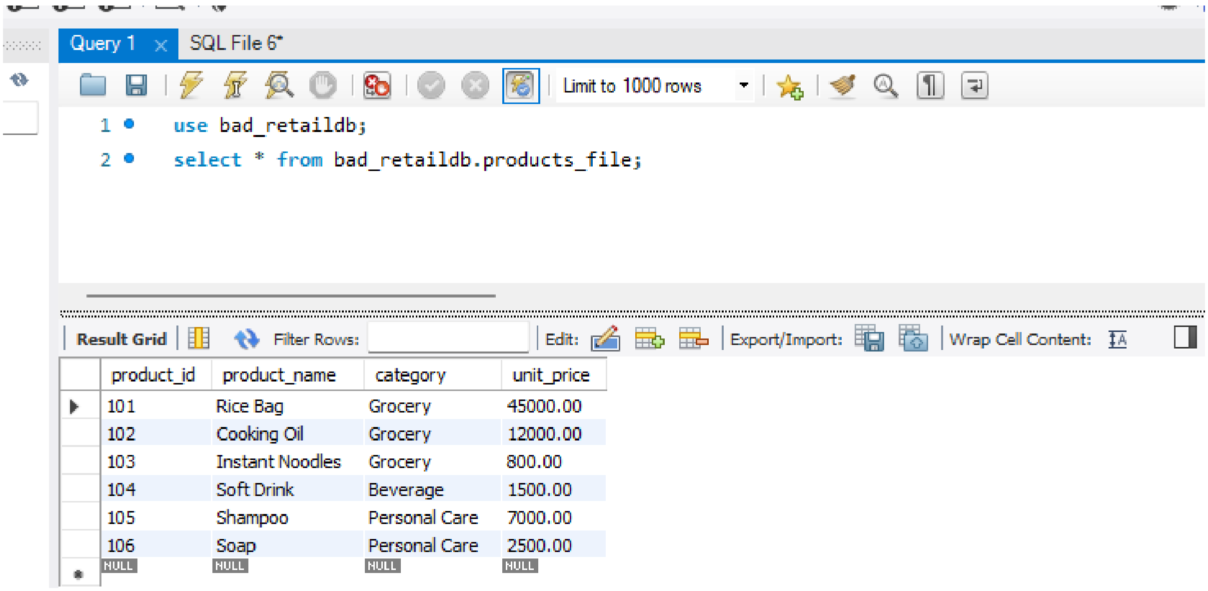
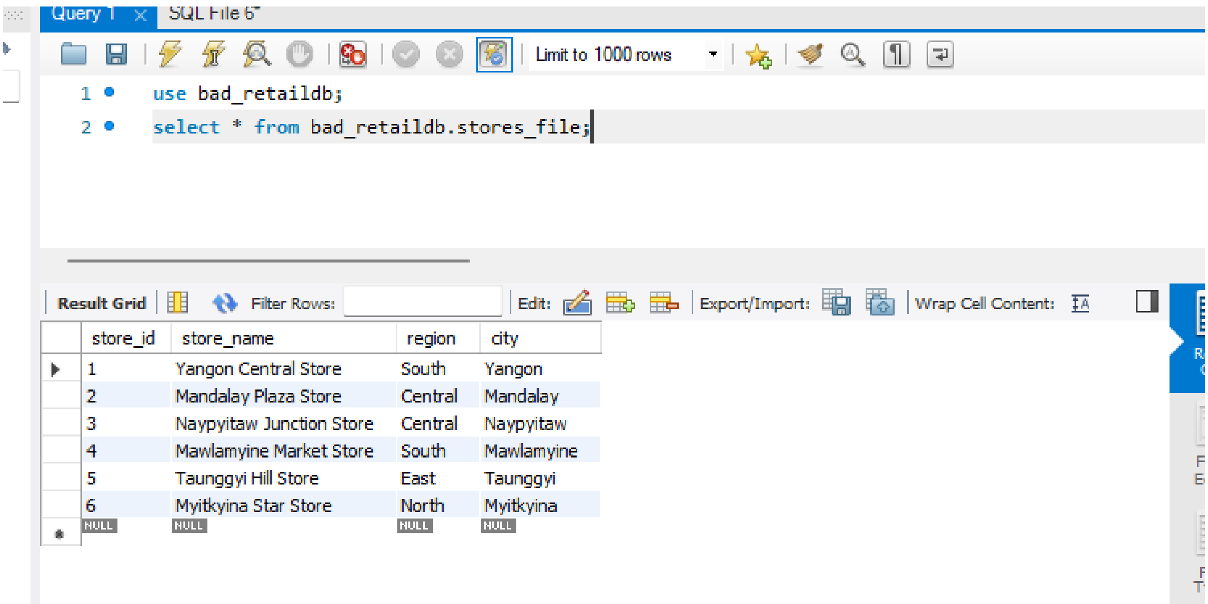
Difficulty in Accessing Data (Retail Store)



```sql
use bad_retaildb;
select * from bad_retaildb.products_file;
```

product_id	product_name	category	unit_price
101	Rice Bag	Grocery	45000.00
102	Cooking Oil	Grocery	12000.00
103	Instant Noodles	Grocery	800.00
104	Soft Drink	Beverage	1500.00
105	Shampoo	Personal Care	7000.00
106	Soap	Personal Care	2500.00
NULL	NULL	NULL	NULL

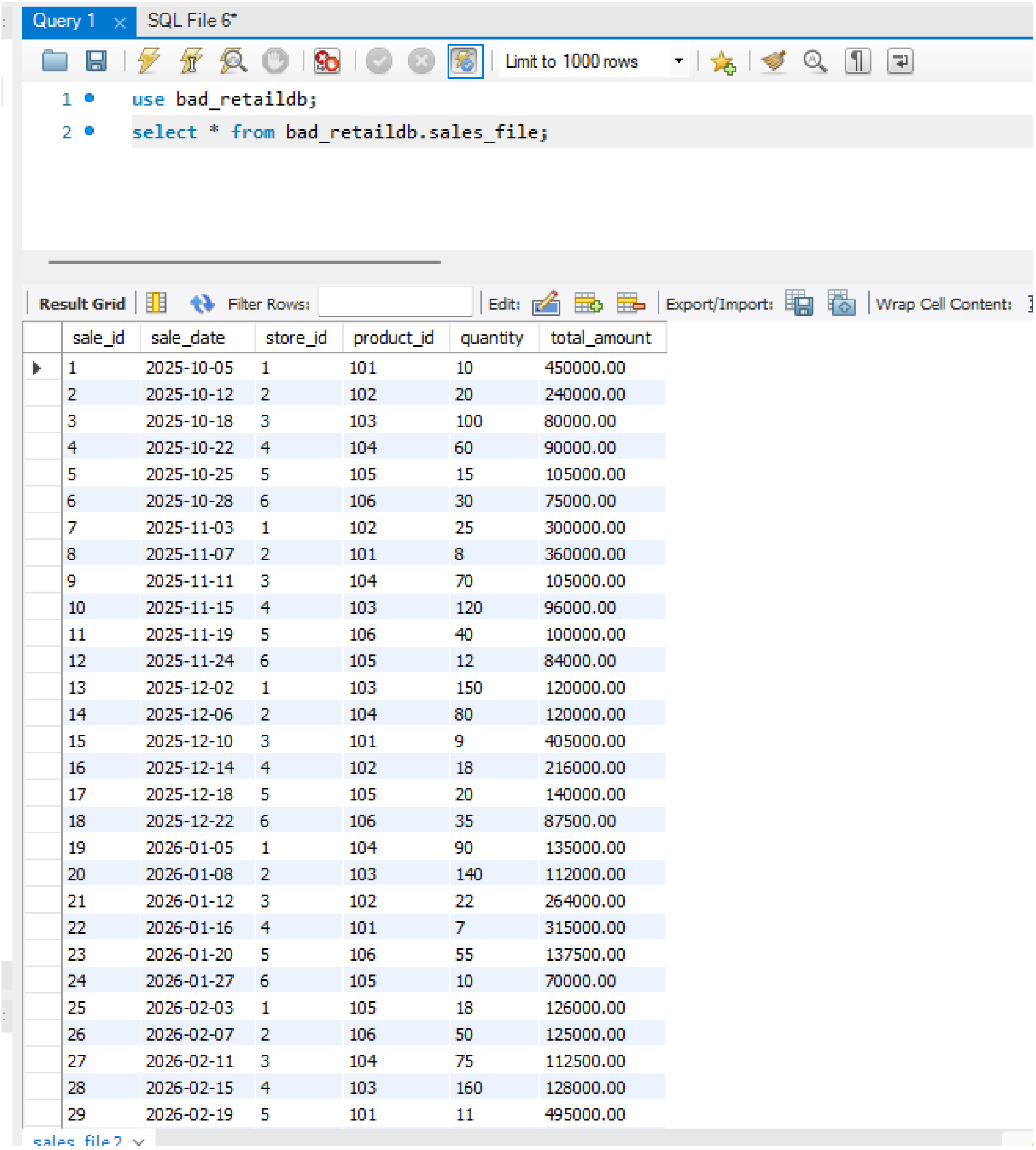
Product_files table



```sql
use bad_retaildb;
select * from bad_retaildb.stores_file;
```

store_id	store_name	region	city
1	Yangon Central Store	South	Yangon
2	Mandalay Plaza Store	Central	Mandalay
3	Naypyitaw Junction Store	Central	Naypyitaw
4	Mawlamyine Market Store	South	Mawlamyine
5	Taunggyi Hill Store	East	Taunggyi
6	Myitkyina Star Store	North	Myitkyina
NULL	NULL	NULL	NULL

Stores_files table



```sql
use bad_retaildb;
select * from bad_retaildb.sales_file;
```

sale_id	sale_date	store_id	product_id	quantity	total_amount
1	2025-10-05	1	101	10	450000.00
2	2025-10-12	2	102	20	240000.00
3	2025-10-18	3	103	100	80000.00
4	2025-10-22	4	104	60	90000.00
5	2025-10-25	5	105	15	105000.00
6	2025-10-28	6	106	30	75000.00
7	2025-11-03	1	102	25	300000.00
8	2025-11-07	2	101	8	360000.00
9	2025-11-11	3	104	70	105000.00
10	2025-11-15	4	103	120	96000.00
11	2025-11-19	5	106	40	100000.00
12	2025-11-24	6	105	12	84000.00
13	2025-12-02	1	103	150	120000.00
14	2025-12-06	2	104	80	120000.00
15	2025-12-10	3	101	9	405000.00
16	2025-12-14	4	102	18	216000.00
17	2025-12-18	5	105	20	140000.00
18	2025-12-22	6	106	35	87500.00
19	2026-01-05	1	104	90	135000.00
20	2026-01-08	2	103	140	112000.00
21	2026-01-12	3	102	22	264000.00
22	2026-01-16	4	101	7	315000.00
23	2026-01-20	5	106	55	137500.00
24	2026-01-27	6	105	10	70000.00
25	2026-02-03	1	105	18	126000.00
26	2026-02-07	2	106	50	125000.00
27	2026-02-11	3	104	75	112500.00
28	2026-02-15	4	103	160	128000.00
29	2026-02-19	5	101	11	495000.00

Sales_files table

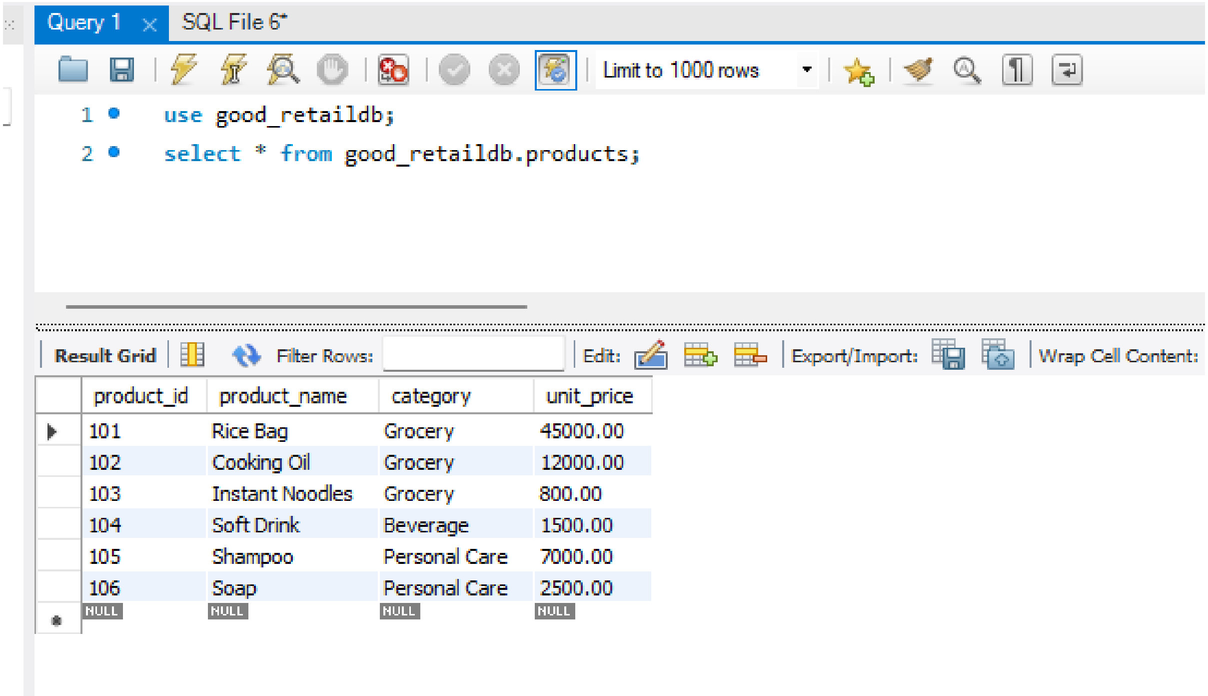
-- Without a proper integrated design, report logic becomes custom and repetitive

```sql
SELECT
    s.sale_date,
    s.store_id,
    st.store_name,
    st.region,
    s.total_amount
FROM sales_file s
JOIN stores_file st ON s.store_id = st.store_id
WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31'
ORDER BY st.region, s.sale_date;
```

sale_date	store_id	store_name	region	total_amount
2025-10-12	2	Mandalay Plaza Store	Central	240000.00
2025-10-18	3	Naypyitaw Junction Store	Central	80000.00
2025-11-07	2	Mandalay Plaza Store	Central	360000.00
2025-11-11	3	Naypyitaw Junction Store	Central	105000.00
2025-12-06	2	Mandalay Plaza Store	Central	120000.00
2025-12-10	3	Naypyitaw Junction Store	Central	405000.00
2026-01-08	2	Mandalay Plaza Store	Central	112000.00
2026-01-12	3	Naypyitaw Junction Store	Central	264000.00
2026-02-07	2	Mandalay Plaza Store	Central	125000.00
2026-02-11	3	Naypyitaw Junction Store	Central	112500.00
2026-03-08	2	Mandalay Plaza Store	Central	98000.00
2026-03-12	3	Naypyitaw Junction Store	Central	144000.00
2025-10-25	5	Taunggyi Hill Store	East	105000.00
2025-11-19	5	Taunggyi Hill Store	East	100000.00
2025-12-18	5	Taunggyi Hill Store	East	140000.00
2026-01-20	5	Taunggyi Hill Store	East	137500.00
2026-02-19	5	Taunggyi Hill Store	East	495000.00

Result 1

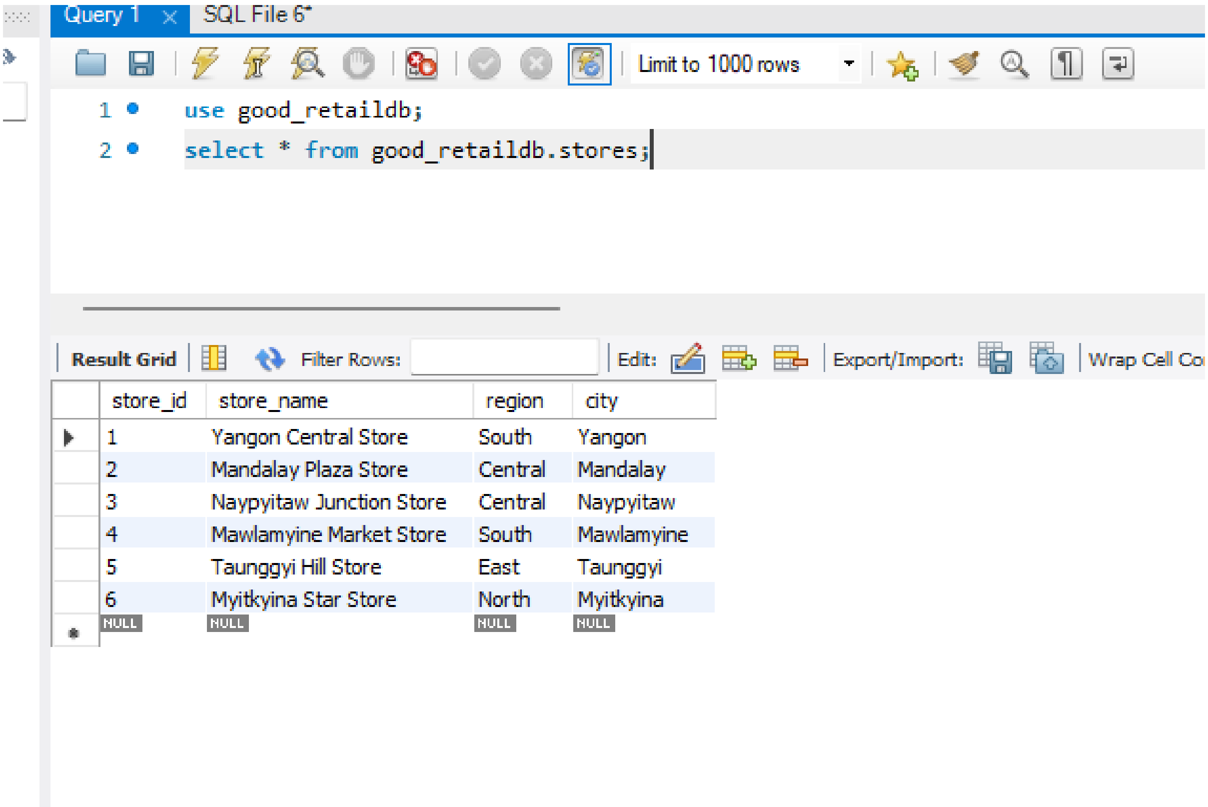
Good retaildb



```
use good_retaildb;
select * from good_retaildb.products;
```

product_id	product_name	category	unit_price
101	Rice Bag	Grocery	45000.00
102	Cooking Oil	Grocery	12000.00
103	Instant Noodles	Grocery	800.00
104	Soft Drink	Beverage	1500.00
105	Shampoo	Personal Care	7000.00
106	Soap	Personal Care	2500.00
NULL	NULL	NULL	NULL

Products table



```
use good_retaildb;
select * from good_retaildb.stores;
```

store_id	store_name	region	city
1	Yangon Central Store	South	Yangon
2	Mandalay Plaza Store	Central	Mandalay
3	Naypyitaw Junction Store	Central	Naypyitaw
4	Mawlamyine Market Store	South	Mawlamyine
5	Taunggyi Hill Store	East	Taunggyi
6	Myitkyina Star Store	North	Myitkyina
NULL	NULL	NULL	NULL

Store table

```sql
use good_retaildb;
select * from good_retaildb.sales;
```

sale_id	sale_date	store_id	product_id	quantity	total_amount
1	2025-10-05	1	101	10	450000.00
2	2025-10-12	2	102	20	240000.00
3	2025-10-18	3	103	100	80000.00
4	2025-10-22	4	104	60	90000.00
5	2025-10-25	5	105	15	105000.00
6	2025-10-28	6	106	30	75000.00
7	2025-11-03	1	102	25	300000.00
8	2025-11-07	2	101	8	360000.00
9	2025-11-11	3	104	70	105000.00
10	2025-11-15	4	103	120	96000.00
11	2025-11-19	5	106	40	100000.00
12	2025-11-24	6	105	12	84000.00
13	2025-12-02	1	103	150	120000.00
14	2025-12-06	2	104	80	120000.00
15	2025-12-10	3	101	9	405000.00
16	2025-12-14	4	102	18	216000.00
17	2025-12-18	5	105	20	140000.00
18	2025-12-22	6	106	35	87500.00
19	2026-01-05	1	104	90	135000.00
20	2026-01-08	2	103	140	112000.00
21	2026-01-12	3	102	22	264000.00
22	2026-01-16	4	101	7	315000.00
23	2026-01-20	5	106	55	137500.00
24	2026-01-27	6	105	10	70000.00
25	2026-02-03	1	105	18	126000.00
26	2026-02-07	2	106	50	125000.00
27	2026-02-11	3	104	75	112500.00
28	2026-02-15	4	103	160	128000.00
29	2026-02-19	5	101	11	495000.00

Sales table

-- Workflow for DBMS Solution

-- the manager’s question can be answered directly by SQL.

-- Sales report for last 6 months by region

```sql
SELECT
    st.region,
    SUM(s.total_amount) AS total_sales
FROM sales s
JOIN stores st ON s.store_id = st.store_id
WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31'
GROUP BY st.region
ORDER BY total_sales DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

region	total_sales
South	2268500.00
Central	2165500.00
East	1229500.00
North	1093500.00

-- Monthly sales report by region

```sql
SELECT
    DATE_FORMAT(s.sale_date, '%Y-%m') AS sales_month,
    st.region,
    SUM(s.total_amount) AS total_sales
FROM sales s
JOIN stores st ON s.store_id = st.store_id
WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31'
GROUP BY DATE_FORMAT(s.sale_date, '%Y-%m'), st.region
ORDER BY sales_month, st.region;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

sales_month	region	total_sales
2025-10	Central	320000.00
2025-10	East	105000.00
2025-10	North	75000.00
2025-10	South	540000.00
2025-11	Central	465000.00
2025-11	East	100000.00
2025-11	North	84000.00
2025-11	South	396000.00
2025-12	Central	525000.00
2025-12	East	140000.00
2025-12	North	87500.00
2025-12	South	336000.00
2026-01	Central	376000.00
2026-01	East	137500.00
2026-01	North	70000.00
2026-01	South	450000.00
2026-02	Central	237500.00
2026-02	East	495000.00
2026-02	North	192000.00
2026-02	South	254000.00

-- Ad-hoc report: sales by region and product category

```sql
-- Ad-hoc report: sales by region and product category
SELECT
    st.region,
    p.category,
    SUM(s.total_amount) AS total_sales
FROM sales s
JOIN stores st ON s.store_id = st.store_id
JOIN products p ON s.product_id = p.product_id
WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31'
GROUP BY st.region, p.category
ORDER BY st.region, total_sales DESC;
-- Create Reporting View
```

sult Grid | Filter Rows: | Export: | Wrap Cell Content:

region	category	total_sales
Central	Grocery	1605000.00
Central	Beverage	337500.00
Central	Personal Care	223000.00
East	Grocery	747000.00
East	Personal Care	482500.00
North	Grocery	777000.00
North	Personal Care	316500.00
South	Grocery	1625000.00
South	Beverage	367500.00
South	Personal Care	276000.00

-- Create Reporting View

```
243 ●   SELECT * FROM vw_sales_reporting;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

sale_id	sale_date	store_id	store_name	region	city	product_id	product_name	category
1	2025-10-05	1	Yangon Central Store	South	Yangon	101	Rice Bag	Grocery
7	2025-11-03	1	Yangon Central Store	South	Yangon	102	Cooking Oil	Grocery
13	2025-12-02	1	Yangon Central Store	South	Yangon	103	Instant Noodles	Grocery
19	2026-01-05	1	Yangon Central Store	South	Yangon	104	Soft Drink	Beverage
25	2026-02-03	1	Yangon Central Store	South	Yangon	105	Shampoo	Personal Care
31	2026-03-04	1	Yangon Central Store	South	Yangon	106	Soap	Personal Care
2	2025-10-12	2	Mandalay Plaza Store	Central	Mandalay	102	Cooking Oil	Grocery
8	2025-11-07	2	Mandalay Plaza Store	Central	Mandalay	101	Rice Bag	Grocery
14	2025-12-06	2	Mandalay Plaza Store	Central	Mandalay	104	Soft Drink	Beverage
20	2026-01-08	2	Mandalay Plaza Store	Central	Mandalay	103	Instant Noodles	Grocery
26	2026-02-07	2	Mandalay Plaza Store	Central	Mandalay	106	Soap	Personal Care
32	2026-03-08	2	Mandalay Plaza Store	Central	Mandalay	105	Shampoo	Personal Care
3	2025-10-18	3	Naypyitaw Junction ...	Central	Naypyitaw	103	Instant Noodles	Grocery
9	2025-11-11	3	Naypyitaw Junction ...	Central	Naypyitaw	104	Soft Drink	Beverage
15	2025-12-10	3	Naypyitaw Junction ...	Central	Naypyitaw	101	Rice Bag	Grocery
21	2026-01-12	3	Naypyitaw Junction ...	Central	Naypyitaw	102	Cooking Oil	Grocery
27	2026-02-11	3	Naypyitaw Junction ...	Central	Naypyitaw	104	Soft Drink	Beverage
33	2026-03-12	3	Naypyitaw Junction ...	Central	Naypyitaw	103	Instant Noodles	Grocery
4	2025-10-22	4	Mawlamyine Market ...	South	Mawlam...	104	Soft Drink	Beverage
10	2025-11-15	4	Mawlamyine Market ...	South	Mawlam...	103	Instant Noodles	Grocery
16	2025-12-14	4	Mawlamyine Market ...	South	Mawlam...	102	Cooking Oil	Grocery
22	2026-01-16	4	Mawlamyine Market ...	South	Mawlam...	101	Rice Bag	Grocery
28	2026-02-15	4	Mawlamyine Market ...	South	Mawlam...	103	Instant Noodles	Grocery
34	2026-03-16	4	Mawlamyine Market ...	South	Mawlam...	104	Soft Drink	Beverage
5	2025-10-25	5	Taunggyi Hill Store	East	Taunggyi	105	Shampoo	Personal Care
11	2025-11-19	5	Taunggyi Hill Store	East	Taunggyi	106	Soap	Personal Care
17	2025-12-18	5	Taunggyi Hill Store	East	Taunggyi	105	Shampoo	Personal Care
23	2026-01-20	5	Taunggyi Hill Store	East	Taunggyi	106	Soap	Personal Care

Vw_sales_reporting table

Test Case

Test Case 1: File-based access is difficult

-- Raw sales file does not directly show region

```sql
USE bad_retaildb;
SELECT
    sale_id,
    sale_date,
    store_id,
    total_amount
FROM sales_file
WHERE sale_date BETWEEN '2025-10-01' AND '2026-03-31';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

sale_id	sale_date	store_id	total_amount
1	2025-10-05	1	450000.00
2	2025-10-12	2	240000.00
3	2025-10-18	3	80000.00
4	2025-10-22	4	90000.00
5	2025-10-25	5	105000.00
6	2025-10-28	6	75000.00
7	2025-11-03	1	300000.00
8	2025-11-07	2	360000.00
9	2025-11-11	3	105000.00
10	2025-11-15	4	96000.00
11	2025-11-19	5	100000.00
12	2025-11-24	6	84000.00
13	2025-12-02	1	120000.00
14	2025-12-06	2	120000.00
15	2025-12-10	3	405000.00
16	2025-12-14	4	216000.00
17	2025-12-18	5	140000.00
18	2025-12-22	6	87500.00
19	2026-01-05	1	135000.00

3Test Case 2: Join is required to retrieve region

```sql
SELECT
    s.sale_id,
    s.sale_date,
    st.region,
    s.total_amount
FROM sales_file s
JOIN stores_file st ON s.store_id = st.store_id
WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

sale_id	sale_date	region	total_amount
1	2025-10-05	South	450000.00
2	2025-10-12	Central	240000.00
3	2025-10-18	Central	80000.00
4	2025-10-22	South	90000.00
5	2025-10-25	East	105000.00
6	2025-10-28	North	75000.00
7	2025-11-03	South	300000.00
8	2025-11-07	Central	360000.00
9	2025-11-11	Central	105000.00
10	2025-11-15	South	96000.00
11	2025-11-19	East	100000.00
12	2025-11-24	North	84000.00
13	2025-12-02	South	120000.00
14	2025-12-06	Central	120000.00
15	2025-12-10	Central	405000.00
16	2025-12-14	South	216000.00
17	2025-12-18	East	140000.00
18	2025-12-22	North	87500.00
19	2026-01-05	South	135000.00

-- Test Case 3: DBMS quickly answers manager's question

```sql
-- Test Case 3: DBMS quickly answers manager’s question
USE good_retaildb;
SELECT
    st.region,
    SUM(s.total_amount) AS total_sales
FROM sales s
JOIN stores st ON s.store_id = st.store_id
WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31'
GROUP BY st.region
ORDER BY total_sales DESC;
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

region	total_sales
South	2268500.00
Central	2165500.00
East	1229500.00
North	1093500.00

-- Test Case 4: Ad-hoc reporting

```sql
SELECT
    region,
    COUNT(*) AS transaction_count,
    SUM(total_amount) AS total_sales
FROM vw_sales_reporting
WHERE sale_date BETWEEN '2025-10-01' AND '2026-03-31'
GROUP BY region
ORDER BY total_sales DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

region	transaction_count	total_sales
South	12	2268500.00
Central	12	2165500.00
East	6	1229500.00
North	6	1093500.00

-- Test Case 5: Monthly trend by region

```sql
SELECT
    DATE_FORMAT(sale_date, '%Y-%m') AS sales_month,
    region,
    SUM(total_amount) AS total_sales
FROM vw_sales_reporting
WHERE sale_date BETWEEN '2025-10-01' AND '2026-03-31'
GROUP BY DATE_FORMAT(sale_date, '%Y-%m'), region
ORDER BY sales_month, region;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

sales_month	region	total_sales
2025-10	Central	320000.00
2025-10	East	105000.00
2025-10	North	75000.00
2025-10	South	540000.00
2025-11	Central	465000.00
2025-11	East	100000.00
2025-11	North	84000.00
2025-11	South	396000.00
2025-12	Central	525000.00
2025-12	East	140000.00
2025-12	North	87500.00
2025-12	South	336000.00
2026-01	Central	376000.00
2026-01	East	137500.00
2026-01	North	70000.00
2026-01	South	450000.00
2026-02	Central	237500.00
2026-02	East	495000.00
2026-02	North	192000.00